



# Modelling Financial Markets

## Chapter 18: Systemic Risk and Networks



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## Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNP: BNP Paribas SA (Euronext Paris ticker; Banque Nationale de Paris)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CATFIN: CATastrophic risk in the FINancial sector
- ▣ CDS: Credit Default Swap
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DCC: Dynamic Conditional Correlation
- ▣ ECB: European Central Bank
- ▣ ES: Expected Shortfall
- ▣ EU: European Union
- ▣ FDIC: Federal Deposit Insurance Corporation
- ▣ FDR: False Discovery Rate
- ▣ FEVD: Forecast Error Variance Decomposition
- ▣ FRM: Financial Risk Meter
- ▣ GAAP: Generally Accepted Accounting Principles (US)
- ▣ GACV: Generalized Approximate Cross-Validation
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HSBC: HSBC Holdings plc (originally the Hongkong and Shanghai Banking Corporation)
- ▣ IFRS: International Financial Reporting Standards
- ▣ ING: ING Groep (Internationale Nederlanden Groep) (the Dutch banking group ING)
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ LRMEs: Long-Run Marginal Expected Shortfall

## Acronyms used in this material (2/2)

- ▣ MES: Marginal Expected Shortfall
- ▣ OLS: Ordinary Least Squares
- ▣ QFVAR: Quantile Factor Vector AutoRegression
- ▣ QML: Quasi-Maximum Likelihood
- ▣ RTD: Relative Tail Dependence
- ▣ SG: Société Générale (the French banking group Société Générale)
- ▣ SRISK: Systemic RISK (capital shortfall measure)
- ▣ SVB: Silicon Valley Bank
- ▣ TENET: Tail-Event driven NETWORK risk
- ▣ US: United States
- ▣ VAR: Vector AutoRegression
- ▣ VIX: CBOE Volatility Index

## Outline

Motivation

Exposure: MES and SRISK

Contribution: CoVaR

Connectedness: Spillover Networks

Tail Networks: the Financial Risk Meter

Climate Risk

Stress Testing

Case Study: Quantile Connectedness

AI for Scientific Discovery

Summary

References

## What Is Systemic Risk?

### From one bank to the system

- ▣ **Systemic risk**: the risk that the distress of one or several institutions impairs the financial system as a whole, and through it the real economy
- ▣ It is an **externality**: a bank does not pay for the losses its failure imposes on others
- ▣ Chapter 7 measured the risk **of** a portfolio; this chapter measures the risk a bank **adds to** or **takes from** the system

### What we do in this chapter

- ▣ Measure the exposure of banks to market crashes: MES and SRISK
- ▣ Measure the contribution of a bank to system losses: CoVaR and  $\Delta\text{CoVaR}$
- ▣ Map the network of volatility spillovers (Diebold–Yilmaz) and tail links (the Financial Risk Meter)
- ▣ Design stress tests: historical, hypothetical and reverse; climate risk

## Bank Runs: Northern Rock 2007 and Silicon Valley Bank 2023



Depositors queue outside a Northern Rock branch,  
Brighton, 14 September 2007

Photo: [Dominic Alves \(2007\)](#); CC BY 2.0; [Wikimedia Commons](#)

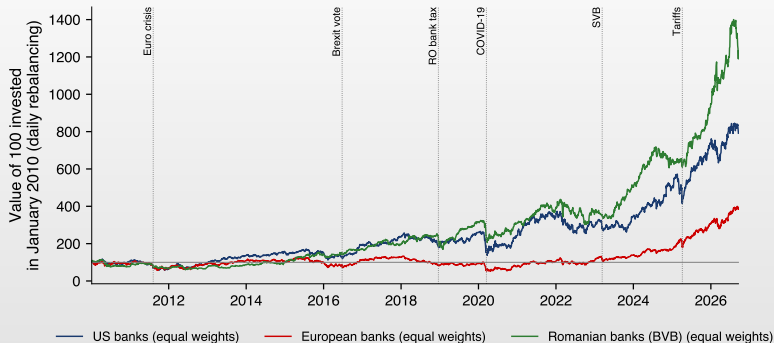
- ❑ Northern Rock (2007): a British mortgage lender hit by the freeze of funding markets after the US subprime losses; depositors queued to withdraw their savings
- ❑ SVB (2023): a digital run; the FDIC took the bank over on 10 March [FDIC \(2023\)](#); two days later all depositors of SVB and Signature Bank were protected on systemic risk grounds [Federal Reserve Board \(2023\)](#)
  - ▶ Section 7 measures the loss of a bank portfolio in that window
- ❑ Channels of contagion
  - ▶ **Funding runs:** depositors and lenders withdraw from any bank that looks similar
  - ▶ **Common exposures and fire sales:** forced selling lowers prices for every holder
  - ▶ **Counterparty links:** one bank's liability is another bank's asset, so a default passes losses to its creditors

## Data Used in This Chapter

| Group          | Series                                                                           | Market index  |
|----------------|----------------------------------------------------------------------------------|---------------|
| US banks       | JPMorgan, Bank of America, Citigroup, Goldman Sachs, Morgan Stanley, Wells Fargo | S&P 500       |
| European banks | Deutsche Bank, BNP Paribas, Santander, ING, HSBC                                 | Euro Stoxx 50 |
| Romanian banks | Banca Transilvania, BRD-Groupe Société Générale                                  | BET           |
| Volatility     | the VIX index (implied volatility of the S&P 500)                                |               |

- Daily prices from January 2010 to 18 September 2026, adjusted for dividends and splits; 3 960 common trading days
- Prices are joined on common days first, then daily log returns  $r_t = 100(\ln P_t - \ln P_{t-1})$ , in %
- Romanian banks: days without trades removed; historical stress scenarios for the international banks start in 2007

## Three Banking Systems, 2010–2026



Equal-weighted portfolios of the banks of each region, rebalanced daily; vertical lines mark the episodes used in this chapter

## Reading the Three Banking Systems

- ▣ 100 invested in January 2010 became 791 in US banks, 384 in European banks and 1233 in Romanian banks
  - ▶ Largest peak-to-trough fall: US 51% (2011-11-23), Europe 61% (2020-04-21), Romania 47% (2012-06-06)
- ▣ Annualised volatility of the portfolios: 29%, 30% and 23%, against 18% (S&P 500), 21% (Euro Stoxx 50) and 18% (BET)
- ▣ The falls are **simultaneous**: the euro crisis, COVID-19 and the 2023 bank failures hit all regions in the same weeks
- ▣ Romanian banks also have their own shock: the bank-tax announcement of December 2018

## Conventions and a Map of Systemic Risk Measures

- Return  $X$ , loss  $L = -X$ ; the level  $\alpha$  is a **tail probability**: VaR 1%, CoVaR 1%, MES 5% (Chapter 7)
  - ▶  $\text{VaR}_\alpha(X) = -q_\alpha(X)$ , with  $q_\alpha$  the  $\alpha$ -quantile; VaR, CoVaR and MES are positive numbers when they are losses
- **System** return  $X^{\text{sys}}$ : the equal-weighted portfolio of the other banks of the same region
- Three families of measures Billio, Getmansky, Lo & Pelizzon (2012)
  - ▶ **Exposure**: how much does the bank lose when the market or the system crashes? MES, SRISK, exposure CoVaR
  - ▶ **Contribution**: how much worse is the system when the bank is in distress?  $\Delta\text{CoVaR}$
  - ▶ **Connectedness**: through which links do shocks travel? Spillover networks, tail networks
- Survey of which measure rests on which theory: Benoit, Colliard, Hurlin & Pérignon (2017); textbooks: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017)

## Recap: Motivation

- ▣ Systemic risk is about the system, not the single bank: it lives in the joint tail of many institutions
- ▣ Runs, common exposures, fire sales and counterparty links turn one failure into many
- ▣ Bank portfolios fall together in crises and are more volatile than the market
- ▣ We need measures of exposure, contribution and connectedness, each with its uncertainty

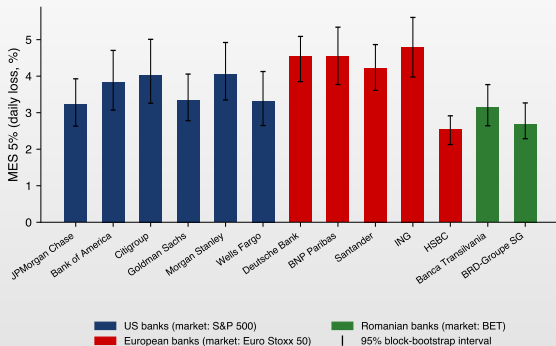
## Marginal Expected Shortfall

- **MES** (Marginal Expected Shortfall) of bank  $i$  at level  $\alpha$  [Acharya, Pedersen, Philippon & Richardson \(2017\)](#)
  - ▶  $\text{MES}_{i,\alpha} = -\mathbb{E}[X_i \mid X_m \leq q_\alpha(X_m)]$
  - ▶ The average loss of the bank on the worst  $\alpha$  share of days for the market  $m$
- Why “marginal”: if the conditioning variable is the portfolio itself,  $m = \sum_i w_i X_i$ , then  $\text{ES}_\alpha(X_m) = \sum_i w_i \text{MES}_{i,\alpha}$  (Euler allocation, next slide)
  - ▶ Our markets (S&P 500, Euro Stoxx 50, BET) are **not** portfolios of these 13 banks: here MES measures **exposure**, not a share of the index ES
- Historical estimate: average  $-X_i$  over the days with  $X_m \leq \hat{q}_\alpha(X_m)$ ; with  $\alpha = 5\%$  and 3 960 days, about 200 days
- Market  $m$ : S&P 500 for US banks, Euro Stoxx 50 for European banks, BET for Romanian banks

## MES as an Euler Contribution: Proof

- Portfolio  $X_p = \sum_i w_i X_i$  with a continuous distribution;  $ES_\alpha(X_p) = -\mathbb{E}[X_p \mid X_p \leq q_\alpha(X_p)]$ 
  - ▶  $ES_\alpha$  is positively homogeneous of degree 1 in  $w$ , so by Euler's theorem  $ES_\alpha(X_p) = \sum_i w_i \partial ES_\alpha / \partial w_i$
  - ▶ Differentiating  $-\frac{1}{\alpha} \mathbb{E}[X_p \mathbf{1}\{X_p \leq q_\alpha(w)\}]$ : the term in  $\partial q_\alpha / \partial w_i$  is  $-\frac{1}{\alpha} q_\alpha f_p(q_\alpha) \partial q_\alpha / \partial w_i + \frac{1}{\alpha} q_\alpha f_p(q_\alpha) \partial q_\alpha / \partial w_i = 0$
  - ▶ Hence  $\partial ES_\alpha / \partial w_i = -\mathbb{E}[X_i \mid X_p \leq q_\alpha(X_p)] = MES_{i,\alpha}$  with  $m = p$  [Gouriéroux, Laurent & Scaillet \(2000\)](#), [Scaillet \(2004\)](#)
- Check on our data: equal-weighted portfolio of the six US banks,  $\alpha = 5\%$  (198 tail days)
  - ▶  $ES_{5\%}$  of the portfolio = 4.32% = average of the six MES conditional on the portfolio (4.32%): exact, also for the historical estimator
  - ▶ Conditioning on the S&P 500 instead gives an average MES of 3.63% (JPMorgan: 3.22% vs 3.85%): the identity fails when  $m \neq p$
- Under the bivariate Normal distribution with zero means,  $MES_{i,\alpha} = \beta_i ES_\alpha(X_m)$ : a beta times a constant (Seminar 18, A3)

## MES 5% of Thirteen Banks



- Average daily loss of each bank on the worst 5% of days for its market index, 2010–2026; 95% intervals from a moving-block bootstrap (blocks of 20 days)

## Reading the MES

- ▣ US banks: from 3.22% (JPMorgan) to 4.04% (Morgan Stanley); European banks up to 4.81% (ING)
  - ▶ On the same days the market lost on average 2.72% (S&P 500) and 3.16% (Euro Stoxx 50): banks lose more than the market
- ▣ HSBC is the exception in Europe (2.54%, below the index): a London listing with Asian business
- ▣ Banca Transilvania 3.14% and BRD 2.69% on BET crash days; BET itself lost 2.70%
- ▣ Overlapping marginal intervals are not a test of equality: bank-to-bank differences need a paired test on joint bootstrap draws (Section 3)

## From MES to a Crisis: LRMES

- **LRMES (Long-Run MES):** the expected fall of the bank's equity in a crisis, a market fall below  $C$  over  $h$  days
  - ▶  $\text{LRMES}_{i,t} = -\mathbb{E}_t [R_{i,t+1:t+h} \mid R_{m,t+1:t+h} < C]$ , arithmetic  $h$ -day returns
  - ▶ [Acharya, Engle & Richardson \(2012\)](#) (Section I):  $C = -40\%$  over six months; [Brownlees & Engle \(2017\)](#) (Section 1.1):  $C = -10\%$  over  $h = 22$  days
- **Shortcut [Acharya, Engle & Richardson \(2012\)](#), Section I**, for “versions of the model where the simulation is not yet implemented”:
 
$$\text{LRMES}_i \approx 1 - \exp(-18 \times \text{MES}_i^{2\%})$$
  - ▶  $\text{MES}_i^{2\%}$ : average daily loss of the bank on days when the market falls by more than 2% (in decimals); example  $\text{MES}^{2\%} = 3\%$  gives  $1 - e^{-0.54} = 41.7\%$
  - ▶ The 18 is a calibration constant; Seminar 18, A4 derives where it comes from
- **Shortcut on our banks:** from 37% (HSBC) to 58% (ING); JPMorgan 48%, Banca Transilvania 52%
  - ▶ The market fell by more than 2% on 138 days in the US, 196 in the euro area and 111 on the BVB

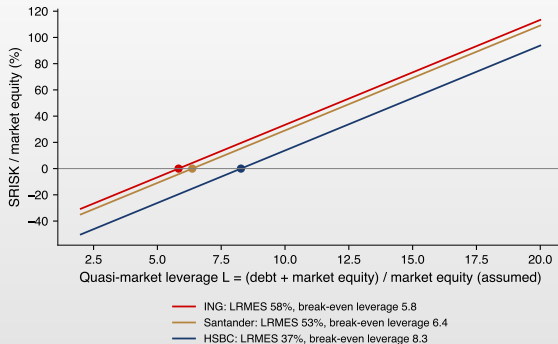
## SRISK: the Capital Shortfall in a Crisis

- A bank must keep equity of at least a share  $k$  of its quasi-assets (book debt  $D$  plus market equity  $W$ ) [Brownlees & Engle \(2017\)](#), Section 1.1, Eq. (1)
  - ▶  $SRISK_i = k D_i - (1 - k)(1 - LRMES_i) W_i$ , assuming debt is not renegotiated in the crisis
  - ▶  $k = 8\%$  for US firms [Brownlees & Engle \(2017\)](#);  $k = 5.5\%$  for European firms, whose IFRS balance sheets report larger assets than US GAAP [Engle, Jondeau & Rockinger \(2015\)](#)
- Positive SRISK = shortfall; the SRISK of the system is the sum of the positive values; [V-Lab, New York University Stern School of Business](#) publishes it for large banks with their balance sheets
- Example:  $D = 900$ ,  $W = 100$ ,  $LRMES = 41.7\%$ 
  - ▶  $SRISK = 0.08 \times 900 - 0.92 \times (1 - 41.7\%) \times 100 = 72 - 53.6 = 18.4$
- SRISK combines a market measure (LRMES) with a balance-sheet measure (leverage)

## SRISK per Unit of Equity and the Break-Even Leverage

- Divide by  $W$  and write the quasi-market leverage  $L = (D + W)/W$ 
  - ▶  $\frac{\text{SRISK}}{W} = k(L - 1) - (1 - k)(1 - \text{LRMES})$
- SRISK is positive once leverage exceeds the **break-even leverage**
  - ▶  $L^* = 1 + \frac{(1 - k)(1 - \text{LRMES})}{k}$
  - ▶ Example above:  $L^* = 7.7$ , while the bank has  $L = 10$ : a shortfall
- Our data contain prices, not balance sheets: we report  $L^*$  from the estimated LRMES and treat leverage as an **assumption**
- A higher LRMES lowers  $L^*$ : the bank can afford less debt before a crisis creates a shortfall

## SRISK as a Function of Leverage



- Banks with the highest, median and lowest LRMES; leverage on the horizontal axis is assumed, not observed; dots mark  $L^*$

## Reading the Break-Even Leverage

- With  $k = 8\%$ :  $L^* = 5.8$  for ING, 6.4 for Santander, 8.3 for HSBC
  - ▶ US: JPMorgan 7.0, Morgan Stanley 6.0; Romania: Banca Transilvania 6.5, BRD 7.1
- A bank whose debt is more than about 5–7 times its market equity would need new capital in a 40% market crash
- With  $k = 5.5\%$  the thresholds rise (ING 8.2, HSBC 11.9): the prudential ratio is a policy choice, and SRISK depends on it
- Limits: LRMES is an extrapolation from daily data, and market equity itself falls in a crisis

## LRMES as Published: GJR–GARCH, DCC and Simulation

- Model of [Brownlees & Engle \(2017\)](#), Section 1.2: zero-mean log returns of bank and market, GJR–GARCH(1,1) volatilities [Glosten, Jagannathan & Runkle \(1993\)](#) and a DCC(1,1) correlation [Engle \(2002\)](#), two-step QML
  - ▶  $\sigma_t^2 = \omega + (\alpha + \gamma \mathbf{1}\{r_{t-1} < 0\}) r_{t-1}^2 + \beta \sigma_{t-1}^2$ , for the bank and for the market
  - ▶  $Q_t = (1 - a - b) S + a \epsilon_{t-1} \epsilon_{t-1}^\top + b Q_{t-1}$ ,  $\rho_t = Q_{12,t} / \sqrt{Q_{11,t} Q_{22,t}}$
- Simulation (Appendix A): resample pairs of standardised innovations  $(\xi_{it}, \epsilon_{mt})$  (next slide), filter  $h$  days ahead from today's  $\sigma^2$  and  $\rho$ , keep the paths with  $R_{m,t+1:t+h} < C$ 
  - ▶ LRMES = minus the average bank return over the crisis paths; resampling pairs keeps the tail dependence that the Normal distribution misses
- Our run: data to 2026-09-18; 200 000 paths of 22 days per market ( $C = -10\%$ ), 400 000 paths of 126 days ( $C = -40\%$ )
  - ▶ Probability of the 22-day crisis today: 1.3% (S&P 500), 4.3% (Euro Stoxx 50), 4.0% (BET); on 31 March 2020: 12%, 19%, 8%

## LRMES Simulation: Innovations and Recursions

- Standardised residuals  $\epsilon_{it} = r_{it}/\sigma_{it}$ ,  $\epsilon_{mt} = r_{mt}/\sigma_{mt}$ ;  $S$  is their sample correlation matrix, the long-run target of  $Q_t$ 
  - ▶ Bank innovation orthogonal to the market:  $\xi_{it} = (\epsilon_{it} - \rho_t \epsilon_{mt})/\sqrt{1 - \rho_t^2}$  [Brownlees & Engle \(2017\)](#), Appendix A
- Each simulated day  $s$ : draw a past date  $\tau$ , take the pair  $(\xi_{i\tau}, \epsilon_{m\tau})$  and rebuild the bank shock with the **simulated** correlation
  - ▶  $\epsilon_{i,t+s} = \rho_{t+s} \epsilon_{m\tau} + \sqrt{1 - \rho_{t+s}^2} \xi_{i\tau}$ , then update  $\sigma^2$  and  $Q$  with the new returns
- Restrictions:  $\omega > 0$ ,  $\alpha \geq 0$ ,  $\alpha + \gamma \geq 0$ ,  $\beta \geq 0$ ,  $\alpha + \gamma/2 + \beta < 1$ ;  $a, b > 0$ ,  $a + b < 1$ , so that  $\sigma_t^2 > 0$  and  $Q_t$  is positive definite
  - ▶ Initialisation:  $Q_1 = S$  in estimation; the simulation starts from  $\sigma_{T+1}^2$  and  $Q_{T+1}$  filtered to the last date

## LRMES by Simulation: Thirteen Banks

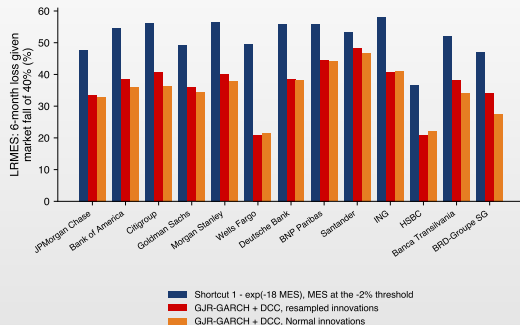
| Bank               | LRMES today | 90% range | LRMES 31.03.2020 | $\rho_T$ | $L^*$ |
|--------------------|-------------|-----------|------------------|----------|-------|
| JPMorgan           | 10.6        | [-2, 25]  | 23.7             | 0.53     | 11.3  |
| Bank of America    | 14.3        | [-4, 35]  | 23.2             | 0.44     | 10.9  |
| Citigroup          | 13.0        | [-1, 28]  | 32.8             | 0.54     | 11.0  |
| Goldman Sachs      | 16.5        | [1, 34]   | 25.0             | 0.65     | 10.6  |
| Morgan Stanley     | 16.3        | [1, 32]   | 27.2             | 0.65     | 10.6  |
| Wells Fargo        | 6.7         | [-8, 24]  | 22.9             | 0.32     | 11.7  |
| Deutsche Bank      | 14.4        | [2, 28]   | 25.7             | 0.66     | 10.8  |
| BNP Paribas        | 15.6        | [2, 33]   | 28.5             | 0.68     | 10.7  |
| Santander          | 16.6        | [3, 34]   | 26.4             | 0.76     | 10.6  |
| ING                | 12.5        | [1, 27]   | 32.9             | 0.68     | 11.1  |
| HSBC               | 8.3         | [-2, 19]  | 10.9             | 0.58     | 11.5  |
| Banca Transilvania | 12.4        | [-2, 34]  | 17.2             | 0.74     | 11.1  |
| BRD-Groupe SG      | 11.4        | [-2, 28]  | 15.8             | 0.64     | 11.2  |

- LRMES (%) for a market fall below  $-10\%$  in 22 days [Brownlees & Engle \(2017\)](#); 90% range: 5% and 95% quantiles of the bank's loss over the crisis paths;  $\rho_T$ : last DCC correlation;  $L^*$  with  $k = 8\%$

## Reading LRMES by Simulation

- LRMES is **conditional**: it moves with today's volatility and correlation
  - ▶ JPMorgan: 10.6% today, 23.7% with the information of 31 March 2020; Deutsche Bank 14.4% and 25.7%
- The 90% range is wide: LRMES is an average over crisis paths, and the capital shortfall of a single crisis can be much larger [Brownlees & Engle \(2017\)](#), Eq. (2)
- The range reflects the crisis scenarios only: parameter uncertainty (GJR and DCC estimates) would widen it further
- With a one-month, -10% crisis, LRMES is about a third of the six-month, -40% numbers of [Acharya, Engle & Richardson \(2012\)](#): the scenario, not only the bank, sets the level

## Model Risk: LRMES under Three Specifications



- Six-month crisis, market fall of 40% Acharya, Engle & Richardson (2012); shortcut  $1 - e^{-18MES^{2\%}}$  vs simulation from GJR-GARCH + DCC with resampled or Normal innovations, data to 2026-09-18

## Reading the Model Risk of LRMES

- Simulated LRMES divided by the shortcut: between 0.42 and 0.90 across the 13 banks; rank correlation 0.77
  - ▶ JPMorgan: 48% (shortcut), 33% (resampled), 33% (Normal innovations); ING: 58%, 41%, 41%
- The shortcut uses the whole 2010–2026 history; the simulation starts from today's volatility and correlation
  - ▶ A 40% fall in six months is rare today: probability 0.29% (S&P 500), 0.96% (Euro Stoxx 50), 0.30% (BET) under the model
- Resampled vs Normal innovations: rank correlation 0.95; the innovation distribution changes levels more than rankings
- Same banks, same data, different defensible models, different numbers: the “model risk of risk models” [Danielsson, James, Valenzuela & Zer \(2016\)](#)

## Recap: MES and SRISK

- MES is the bank's average loss on the market's worst days; weighted by portfolio shares, the MES add up to the ES of a portfolio only when they condition on that same portfolio
- LRMES extrapolates MES to a six-month crisis; SRISK turns it into the capital shortfall
- European banks have the largest MES in our sample; bank and region rankings carry wide intervals
- As published, LRMES is simulated from GJR-GARCH + DCC; the  $1 - e^{-18\text{MES}}$  shortcut differs by bank, and the scenario ( $h$ ,  $C$ ) sets the level
- SRISK needs balance-sheet data; without it we can report only the break-even leverage

## CoVaR: the VaR of the System Given a Bank in Distress

- **CoVaR** of the system conditional on bank  $i$  at its VaR [Adrian & Brunnermeier \(2016\)](#)
  - ▶  $P\left(X^{sys} \leq -\text{CoVaR}_{\alpha}^{sys|i} \mid X_i = -\text{VaR}_{i,\alpha}\right) = \alpha$
  - ▶ The loss of the system that is exceeded with probability  $\alpha$  on a day when bank  $i$  is at its own VaR
- **$\Delta\text{CoVaR}$** : the change from the median state of the bank to its distress state
  - ▶  $\Delta\text{CoVaR}_{\alpha}^{sys|i} = \text{CoVaR}_{\alpha}^{sys|X_i=q_{\alpha}} - \text{CoVaR}_{\alpha}^{sys|X_i=q_{50\%}}$
- **$\Delta\text{CoVaR}$**  measures the **contribution** of bank  $i$  to the tail risk of the system
- Reversing the roles gives the **exposure CoVaR**: the VaR of bank  $i$  when the system is in distress

## Quantile Regression

- The  $\alpha$ -quantile of  $Y$  given  $x$  is linear:  $q_\alpha(Y | x) = a_\alpha + b_\alpha x$  [Koenker & Bassett \(1978\)](#)
  - ▶  $(\hat{a}_\alpha, \hat{b}_\alpha) = \arg \min_{a,b} \sum_t \rho_\alpha(y_t - a - b x_t)$ ,  $\rho_\alpha(u) = u(\alpha - 1\{u < 0\})$
  - ▶  $\rho_\alpha$  is the **check function**: it weights negative residuals by  $1 - \alpha$  and positive ones by  $\alpha$ ;  $\alpha = 50\%$  gives the median regression
- Asymptotics for a fixed  $\alpha$ , with  $x_t = (1, X_{i,t})^\top$  and  $f_t$  the conditional density of  $Y$  at its  $\alpha$ -quantile
  - ▶  $\sqrt{n}(\hat{\theta}_\alpha - \theta_\alpha) \Rightarrow N\left(0, \alpha(1 - \alpha) D_1^{-1} D_0 D_1^{-1}\right)$ ,  $D_0 = \mathbb{E}[x_t x_t^\top]$ ,  $D_1 = \mathbb{E}[f_t x_t x_t^\top]$
  - ▶ Assumes independent observations (or scores  $x_t(\alpha - 1\{u_t < 0\})$  that are a martingale difference),  $f_t > 0$  near the quantile and  $D_0, D_1$  positive definite; with serially dependent scores  $D_0$  needs a long-run (HAC) estimate or a block bootstrap
  - ▶ The **sparsity**  $1/f_t$  must be estimated (kernel or bandwidth); at  $\alpha = 1\%$  few observations lie near the quantile, so this estimate is noisy
- The slope  $b_\alpha$  can change with  $\alpha$ : the bank may matter more for the system in the tail than in the centre; least squares gives only the mean slope

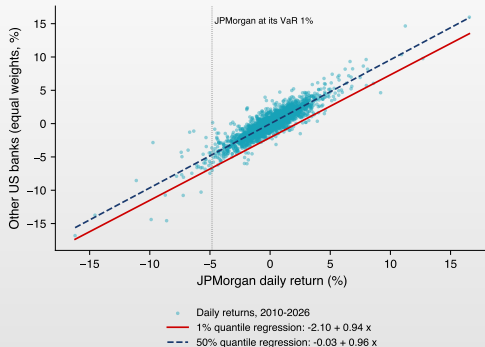
## Estimating CoVaR in Three Steps

- ▣ 1. Quantile regression of the system on the bank at level  $\alpha$ 
  - ▶  $\hat{q}_\alpha(X^{sys} | X_i) = \hat{a}_\alpha + \hat{b}_\alpha X_i$
- ▣ 2. Plug in the bank's own quantiles:  $\hat{q}_\alpha(X_i)$  (distress) and  $\hat{q}_{50\%}(X_i)$  (median)
  - ▶  $\text{CoVaR}_\alpha^{sys|i} = - \left( \hat{a}_\alpha + \hat{b}_\alpha \hat{q}_\alpha(X_i) \right)$
- ▣ 3. Take the difference
  - ▶  $\Delta \text{CoVaR}_\alpha^{sys|i} = \hat{b}_\alpha (\hat{q}_{50\%}(X_i) - \hat{q}_\alpha(X_i))$
  - ▶ Only the slope matters: the intercept cancels
- ▣ Uncertainty: moving-block bootstrap of the days (blocks of 20 days keep volatility clusters together), 999 resamples

## Distress at or below VaR: the Girardi–Ergün CoVaR

- Girardi & Ergün (2013) condition on the bank being at most at its VaR, not exactly at it
  - ▶  $P\left(X^{sys} \leq -\text{CoVaR}_{\alpha}^{\leq} \mid X_i \leq -\text{VaR}_{i,\alpha}\right) = \alpha \iff P\left(X^{sys} \leq -\text{CoVaR}_{\alpha}^{\leq}, X_i \leq -\text{VaR}_{i,\alpha}\right) = \alpha^2$
  - ▶ Benchmark state: a one-standard-deviation event  $|X_i - \mu_i| \leq \sigma_i$  instead of the median
- Why it matters
  - ▶ It includes more severe distress than the VaR day itself, and it is monotone in the dependence parameter
  - ▶ It can be backtested: the joint hit  $\{X^{sys} \leq -\text{CoVaR}^{\leq}, X_i \leq -\text{VaR}_i\}$  has probability  $\alpha^2$
- Historical estimate at  $\alpha = 5\%$ : the 5% quantile of the system on the 198 days with  $X_i \leq \hat{q}_{5\%}(X_i)$ 
  - ▶ JPMorgan:  $\text{CoVaR}_{5\%}^{\leq} = 7.70\%$  vs the quantile-regression  $\text{CoVaR}_{5\%} = 3.81\%$ ; eleven banks: 7.3–8.2% vs 3.6–4.8%
  - ▶ About twice as large: averaging over the whole tail of the bank is a harsher conditioning event (Seminar 18, A2 derives this under the Normal distribution)

## Quantile Regression: JPMorgan and the Other US Banks

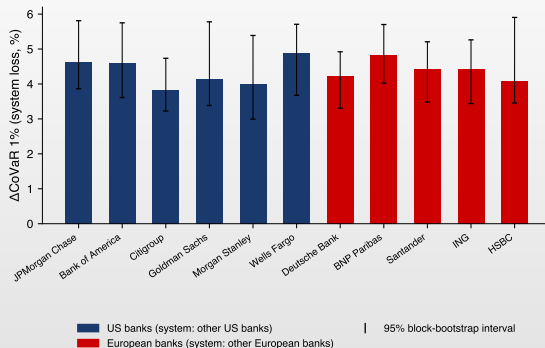


- Daily returns 2010–2026 with the 1% and 50% quantile regression lines; the dotted line marks JPMorgan at its VaR 1%

## Reading the JPMorgan Example

- ▣ 1% line:  $\hat{q}_{1\%} = -2.10 + 0.94 X_i$ ; median line:  $-0.03 + 0.96 X_i$ 
  - ▶ Similar slopes: the 1% line is shifted down by the tail of the other banks, not steeper
- ▣ JPMorgan's VaR 1% is 4.83%; on such a day the other US banks have CoVaR 1% = 6.64%
  - ▶ Unconditional VaR 1% of the system: 5.07%; with JPMorgan at its median: 2.03%
- ▣  $\Delta\text{CoVaR}_{1\%} = 4.62\%$ : JPMorgan in distress adds about 4.62 percentage points to the 1% loss of the other US banks
- ▣ This is association, not causation: a common shock hits JPMorgan and the others on the same day

## $\Delta\text{CoVaR}$ 1% of Eleven Banks



□ System: equal-weighted portfolio of the other banks of the same region; 95% moving-block bootstrap intervals

## Reading $\Delta\text{CoVaR}$

- All eleven banks lie between 3.81% (Citigroup) and 4.89% (Wells Fargo); the 95% intervals are 1.5–2.4 points wide
  - ▶ E.g. Citigroup [3.23, 4.74], Wells Fargo [3.68, 5.71]: the marginal intervals overlap
- Overlap is **not** a test: two estimates computed on the same days are dependent (in our bootstrap draws strongly positively correlated), so the difference can be tighter than either interval
  - ▶ The next two slides test the differences on joint bootstrap draws
- At  $\alpha = 5\%$  the values are smaller (JPMorgan 2.62%) and the ranking changes: Spearman correlation between the 1% and 5% rankings 0.10
  - ▶ A ranking that changes with  $\alpha$  is not a robust basis for regulation

## Testing $\Delta\text{CoVaR}$ : Three Hypotheses

- (1) No contribution,  $H_0: b_\alpha = 0$ : Wald test with the sandwich variance of the previous slides [Koenker & Bassett \(1978\)](#)
  - ▶ Tests of linear hypotheses across quantiles use the joint asymptotics of  $(\hat{b}_\alpha, \hat{b}_{50\%})$  [Koenker & Bassett \(1982\)](#)
- (2) The tail link equals the central link,  $H_0: b_\alpha = b_{50\%}$ : is the bank special in crises?
  - ▶ We estimate the variance of  $\hat{b}_\alpha - \hat{b}_{50\%}$  from 999 moving-block bootstrap draws that keep the pairs of days together
- (3) Equal contributions,  $H_0: \Delta\text{CoVaR}_i = \Delta\text{CoVaR}_j$ : the same resampled days for **all** banks give the joint distribution
  - ▶  $z_{ij} = (\widehat{\Delta\text{CoVaR}}_i - \widehat{\Delta\text{CoVaR}}_j) / \widehat{\text{se}}^*(\cdot)$  for each pair; Holm correction over the pairs [Holm \(1979\)](#); a joint Wald test that all are equal
  - ▶ Significance and dominance tests for systemic importance: [Castro & Ferrari \(2014\)](#)

Testing  $\Delta\text{CoVaR}$  1%: Results

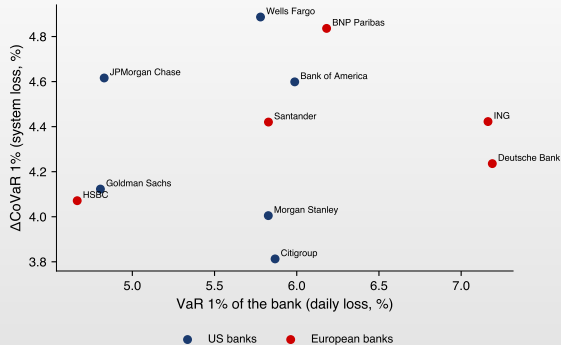
| Bank            | $\Delta\text{CoVaR}$ | $\hat{b}_{1\%}$ | $z$ | $\hat{b}_{50\%}$ | $\hat{b}_{1\%} - \hat{b}_{50\%}$ | $p$  |
|-----------------|----------------------|-----------------|-----|------------------|----------------------------------|------|
| JPMorgan        | 4.62                 | 0.94            | 6.9 | 0.96             | -0.02                            | 0.66 |
| Bank of America | 4.60                 | 0.76            | 6.5 | 0.78             | -0.02                            | 0.73 |
| Citigroup       | 3.81                 | 0.64            | 4.7 | 0.76             | -0.11                            | 0.00 |
| Goldman Sachs   | 4.12                 | 0.85            | 5.6 | 0.84             | 0.00                             | 0.99 |
| Morgan Stanley  | 4.00                 | 0.68            | 4.5 | 0.73             | -0.05                            | 0.28 |
| Wells Fargo     | 4.89                 | 0.84            | 4.3 | 0.78             | 0.06                             | 0.17 |
| Deutsche Bank   | 4.24                 | 0.59            | 4.1 | 0.60             | -0.02                            | 0.67 |
| BNP Paribas     | 4.84                 | 0.77            | 6.5 | 0.74             | 0.03                             | 0.37 |
| Santander       | 4.42                 | 0.75            | 4.5 | 0.69             | 0.07                             | 0.07 |
| ING             | 4.42                 | 0.61            | 4.4 | 0.71             | -0.10                            | 0.00 |
| HSBC            | 4.07                 | 0.87            | 4.0 | 0.86             | 0.01                             | 0.96 |

- $z$ : kernel (sandwich)  $t$ -ratio of  $\hat{b}_{1\%}$ , valid only for serially independent scores: every bank contributes;  $p$ : bootstrap test of  $b_{1\%} = b_{50\%}$ , significant for 2 of 11 banks (2 after Holm)
- ▶ Equal contributions within a region: 4 of 25 pairs differ at 5%, 0 after Holm; joint Wald  $p = 0.01$  (US), 0.31 (Europe)
  - ▶ At 5%: 3 of 25 pairs differ, 0 after Holm; joint Wald  $p = 0.014$  (US), 0.046 (Europe)

## Extremal Quantiles: Why $\alpha = 1\%$ Is Hard

- The effective tail sample is  $\alpha n$ : 40 days at 1% and 198 at 5% out of 3 960
  - ▶ The Normal approximation above assumes  $\alpha n \rightarrow \infty$  fast; with  $\alpha n$  of a few dozen it can be poor, and extremal-quantile theory (extreme-value limits, subsampling) is the alternative built for this case [Chernozhukov & Fernández-Val \(2011\)](#)
- In our data the block-bootstrap s.e. of  $\Delta\text{CoVaR } 1\%$  is about 0.6 times the kernel-based s.e. (median over banks): the two routes disagree by a large factor
  - ▶ The kernel s.e. rests on a sparsity estimate from the few observations near the 1% quantile; the bootstrap redraws the same few tail days: neither is reliable when  $\alpha n$  is this small
- Practical rule: report  $\alpha = 5\%$  and  $\alpha = 1\%$  with  $\alpha n$ ; expect the 1% ranking to be the noisier one
  - ▶ The low Spearman correlation between the 1% and 5% rankings (0.10) may reflect estimation noise, genuine differences between the two tail levels, or both

## VaR and $\Delta\text{CoVaR}$ Measure Different Things



□ Horizontal: VaR 1% of the bank; vertical: its  $\Delta\text{CoVaR}$  1%, eleven banks, 2010–2026

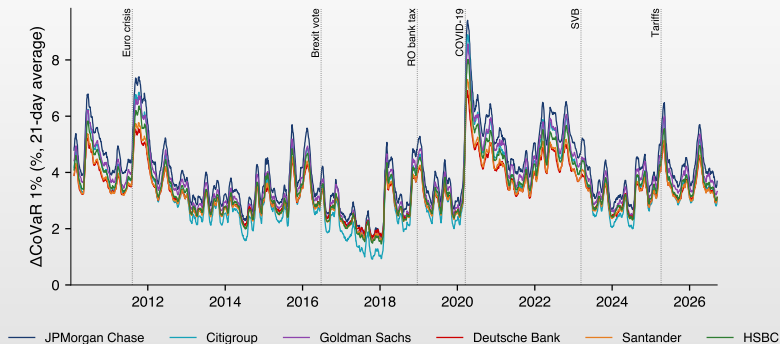
## Reading VaR versus $\Delta\text{CoVaR}$

- Spearman rank correlation between VaR and  $\Delta\text{CoVaR}$ : 0.16
  - ▶ Deutsche Bank and ING have the largest VaR (7.19%, 7.16%) but only average  $\Delta\text{CoVaR}$
  - ▶ JPMorgan has one of the smallest VaR values (4.83%) and one of the largest  $\Delta\text{CoVaR}$  values
- This is the central message of [Adrian & Brunnermeier \(2016\)](#): regulating each bank by its own VaR ignores its contribution to the system
- Caveat: with eleven banks and overlapping intervals, the correlation itself is very uncertain (Seminar 18, B2)

## Time-Varying $\Delta\text{CoVaR}$ with State Variables

- State variables  $M_{t-1}$ : log VIX, the return of the regional index and of the system, all lagged one day [Adrian & Brunnermeier \(2016\)](#)
  - ▶  $q_\alpha(X_{i,t} \mid M_{t-1}) = a_i + c_i^\top M_{t-1}$  at  $\alpha$  and at 50%
  - ▶  $q_\alpha(X_t^{\text{sys}} \mid X_{i,t}, M_{t-1}) = a + b X_{i,t} + c^\top M_{t-1}$
- Then  $\Delta\text{CoVaR}_{\alpha,t} = \hat{b} (\hat{q}_{50\%,t}(X_i) - \hat{q}_{\alpha,t}(X_i))$ 
  - ▶ It moves with the bank's conditional VaR, which moves with the VIX
- The slope  $\hat{b}$  is constant: time variation comes from the state, not from the link

## $\Delta\text{CoVaR 1\%}$ over Time

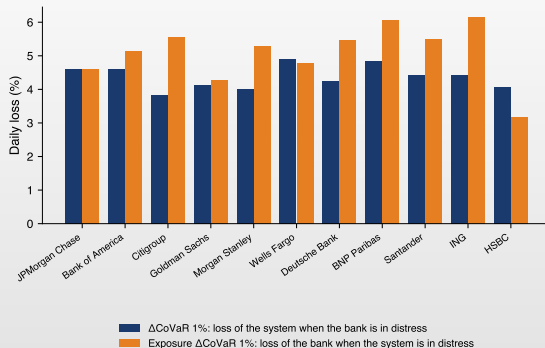


- Six banks, 21-day moving averages of the daily  $\Delta\text{CoVaR 1\%}$  estimated with lagged state variables

## Reading $\Delta\text{CoVaR}$ over Time

- ▣ JPMorgan: 2.16% on average in 2017, 9.06% in March 2020; peak 10.99% on 2020-03-18
  - ▶ Deutsche Bank: 2.04% in 2017 and 6.65% in March 2020
- ▣ The six lines move together: the common state variables drive them
- ▣ Market-based  $\Delta\text{CoVaR}$  is low in calm periods, exactly when risk may be building up in balance sheets, which market data do not see
- ▣ [Adrian & Brunnermeier \(2016\)](#) therefore propose a forward  $\Delta\text{CoVaR}$ , predicted from leverage, size and maturity mismatch

## Contribution versus Exposure



- Blue:  $\Delta\text{CoVaR 1\%}$  (loss of the system when the bank is in distress); orange: exposure  $\Delta\text{CoVaR 1\%}$  (loss of the bank when the system is in distress)

## Reading Contribution versus Exposure

- The direction matters: the two numbers differ for most banks
  - ▶ ING: contribution 4.42%, exposure 6.14%; BNP Paribas: 4.84% and 6.07%
  - ▶ HSBC: contribution 4.07%, exposure only 3.17%
- Exposure is larger than contribution for most European banks: the rise in the bank's tail loss when the system is in distress exceeds the rise in the system's tail loss when the bank is in distress; both are associations, not causal effects
- Exposure matters for the bank's own capital; contribution matters for the regulator of the system

## Backtesting CoVaR and MES

- Systemic measures are conditional risk measures, so they need **conditional** backtests Banulescu-Radu, Hurlin, Leymarie & Scaillet (2021)
  - ▶ For  $\text{CoVaR}^{\leq}$ : hit  $I_t = 1\{X_t^{\text{sys}} \leq -\text{CoVaR}_t^{\leq}, X_{i,t} \leq -\text{VaR}_{i,t}\}$  with  $P(I_t = 1) = \alpha^2$  under  $H_0$  if both the bank VaR and the conditional quantile are calibrated; Kupiec and Christoffersen tests on  $I_t$  (Chapter 8), plus a separate check of the bank VaR rate
  - ▶ For MES: the cumulative joint violation process of Banulescu-Radu, Hurlin, Leymarie & Scaillet (2021)
- Our out-of-sample check: historical  $\text{CoVaR}_{5\%}^{\leq}$  on an expanding window from day 1 000, 2 960 forecast days per bank
  - ▶ Expected joint hits 7.4; observed 5–9 across the eleven banks (JPMorgan 9); smallest Kupiec  $p = 0.35$ : no rejection
  - ▶ Power is limited: if the true joint rate were twice  $\alpha^2$ , the 5% Kupiec test would reject in 62% of samples of this length with independent hits, so even a doubling is missed about one time in three

## How Much Can We Trust a Ranking?

- Rankings of banks by different systemic measures disagree, and each ranking has wide sampling uncertainty [Nucera, Schwaab, Koopman & Lucas \(2016\)](#)
  - ▶ They propose combining measures and reporting rank uncertainty instead of a single ordering
- Model risk: the same bank's risk measure varies widely across estimation methods [Danielsson, James, Valenzuela & Zer \(2016\)](#)
  - ▶ In this chapter: LRMES shortcut vs simulation (ratio 0.42–0.90), quantile-regression vs Girardi–Ergün CoVaR (about  $\times 2$ ),  $\alpha = 1\%$  vs  $5\%$  (Spearman 0.10)
- What survives: large, tested differences (regions; the tail slope of some banks), not the order of eleven close point estimates

## Recap: CoVaR

- ▣ CoVaR is the VaR of the system given a bank in distress;  $\Delta\text{CoVaR}$  is the bank's contribution
- ▣ Estimated by quantile regression; only the tail slope  $\hat{b}_\alpha$  and the bank's quantiles matter
- ▣ VaR and  $\Delta\text{CoVaR}$  rank banks differently; differences between banks must be tested on joint draws, with a multiplicity correction
- ▣ With state variables,  $\Delta\text{CoVaR}$  follows market volatility: it rises in crises and says little in calm times

## From Pairs to Networks

- CoVaR looks at one bank and one system; a **network** looks at all pairs at once
  - ▶ Nodes: banks; directed edges: how much of the future uncertainty of bank  $i$  comes from shocks to bank  $j$
- Granger-causality networks of banks, insurers, hedge funds and brokers Billio, Getmansky, Lo & Pelizzon (2012)
- Variance-decomposition networks Diebold & Yilmaz (2009), Diebold & Yilmaz (2012), Diebold & Yilmaz (2014): the approach of this section
- Tail networks from quantile regressions: TENET Härdle, Wang & Yu (2016)

## The Forecast Error Variance Decomposition

- Vector autoregression (VAR) of order  $p$  for  $N$  series  $y_t$ :  $y_t = c + \sum_{l=1}^p A_l y_{t-l} + \varepsilon_t$ ,  $\text{Cov}(\varepsilon_t) = \Sigma$ 
  - ▶ Moving-average form:  $y_t = \mu + \sum_{h=0} \Phi_h \varepsilon_{t-h}$ ,  $\Phi_0 = I$ ,  $\Phi_h = \sum_{l=1}^p A_l \Phi_{h-l}$
- The  $H$ -day forecast error of  $y_i$  is a sum of shocks to all series: its variance can be split by source
- Orthogonal (Cholesky) shocks depend on the ordering of the series; the **generalized** decomposition does not [Koop, Pesaran & Potter \(1996\)](#), [Pesaran & Shin \(1998\)](#)
  - ▶  $\theta_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i^\top \Phi_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} e_i^\top \Phi_h \Sigma \Phi_h^\top e_i}$ , normalised:  $\tilde{\theta}_{ij} = \theta_{ij} / \sum_j \theta_{ij}$

## The Connectedness Table

- $\tilde{\theta}_{ij}$  (in %): the share of the  $H$ -day forecast error variance of  $i$  due to shocks in  $j$ ; each row sums to 100 [Diebold & Yilmaz \(2014\)](#)
  - ▶ **From others:**  $F_i = \sum_{j \neq i} \tilde{\theta}_{ij}$ ; **to others:**  $T_j = \sum_{i \neq j} \tilde{\theta}_{ij}$
  - ▶ **Net:**  $T_i - F_i$  (positive = net transmitter); **pairwise net:**  $\tilde{\theta}_{ji} - \tilde{\theta}_{ij}$
- **Total connectedness:**  $C = \frac{1}{N} \sum_{i \neq j} \tilde{\theta}_{ij}$ , the average share of variance that comes from others
  - ▶ Rows of  $\theta_{ij}$  do not sum to one before normalisation because generalized shocks are correlated (Seminar 18, A5 derives this)
  - ▶ A pairwise net link  $\tilde{\theta}_{ji} - \tilde{\theta}_{ij} > 0$  means that  $i$  transmits net to  $j$  and draws an arrow  $i \rightarrow j$ : the network of the next slides

## What We Connect: Daily Range Volatility

- **Parkinson volatility** from the daily high  $H_t$  and low  $L_t$  [Parkinson \(1980\)](#)

- ▶  $\hat{\sigma}_t^2 = \frac{(\ln H_t - \ln L_t)^2}{4 \ln 2}$ , annualised  $100\sqrt{252 \hat{\sigma}_t^2}$ , in logs

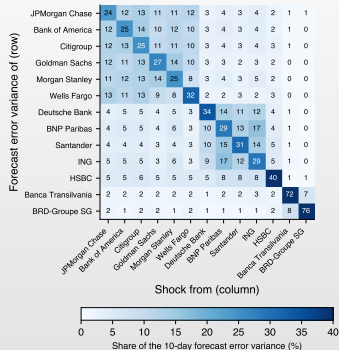
- ▶ Unbiased for the variance of a Brownian motion without drift; more efficient than the squared return

- Volatility is persistent and therefore forecastable, while daily returns are close to unpredictable: a natural object for a VAR

- Our model: VAR on the log volatility of 13 banks, order chosen by the Bayesian information criterion ( $p = 2$ ), horizon  $H = 10$  days

- ▶ Time zones: US banks close after European ones; the VAR lags capture the next-day transmission

## The Connectedness Table of Thirteen Banks

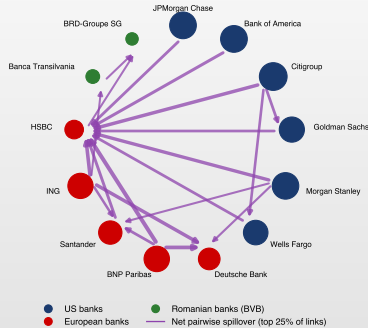


□ Row  $i$ , column  $j$ : share (%) of the 10-day forecast error variance of bank  $i$  due to shocks in bank  $j$ , full sample

## Reading the Connectedness Table

- Total connectedness  $C = 63.9\%$ : on average two thirds of a bank's volatility risk comes from the other banks
  - ▶ On returns instead of volatility:  $C = 75.4\%$
- Two regional blocks: a US bank receives 84.0% from US shocks and a European bank 72.7% from European shocks, own shock included
  - ▶ Across the Atlantic: a European bank receives 26.2% from US banks, a US bank 15.0% from European ones
- Romanian banks are largely their own: own share 72.3% (Banca Transilvania) and 76.1% (BRD); they receive 10.3% from US and 8.0% from European banks

## The Net Spillover Network



- Arrow from  $j$  to  $i$  when  $j$  transmits net to  $i$  (largest 25% of pairwise net spillovers); node size grows with the spillover to others

## Reading the Network

- Net transmitters: Citigroup (+19.0), Bank of America (+10.9), Morgan Stanley (+9.9), JPMorgan (+9.8)
  - ▶ To others: Citigroup 93.8%, JPMorgan 86.2% (the sum of a column can exceed 100)
- Net receivers: HSBC (−23.6), BRD (−13.0), Deutsche Bank (−12.6), Banca Transilvania (−12.2)
  - ▶ HSBC receives from almost every bank: it trades in London, between Asia and New York
- Within Europe, BNP Paribas and ING transmit to Deutsche Bank and Santander
- A full-sample network averages quiet and crisis years: the next chart lets it move

## Connectedness over Time



- Top: total connectedness on moving 250-day windows ( $\text{VAR}(2)$ ,  $H = 10$ , every 5 days); bottom: net spillover summed by region

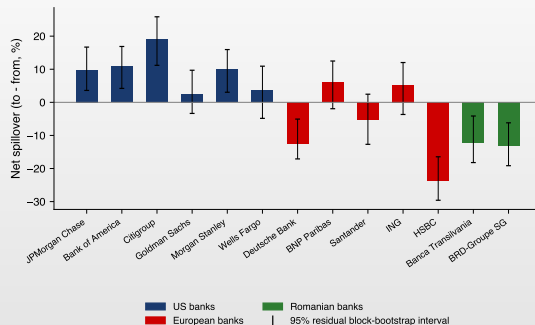
## Reading Connectedness over Time

- ▣ The total index ranges from 47.4% (2024-07-24) to 79.3% (2020-08-27); mean 62.0%
  - ▶ COVID-19: from 61.8% in 2019 to 78.4% in spring 2020; it jumps within days and decays over a year
  - ▶ After the 2023 bank failures 66.0%; the calm year 2024 53.5%
- ▣ US banks are net transmitters in 89% of the windows (average net +48.4)
  - ▶ European banks were net transmitters during the euro crisis of 2011–2012 and mostly net receivers afterwards
- ▣ Romanian banks are net receivers on average (−24.1)
- ▣ Lag order:  $p = 2$  is held fixed in every window so that the index does not jump when the selected order changes; BIC picks  $p = 1$  in 100% of the windows, and with  $p = 1$  the index has correlation 0.99 with the one shown

## Estimation Error in the Network

- A VAR(1) on 13 series has 169 slope coefficients; a 250-day window gives 250 observations per equation for 13 slopes, and the table is a nonlinear function of all of them
  - ▶ Every number in the table and every arrow of the network is an estimate: it needs an interval
- Residual block bootstrap: resample blocks of 20 days of VAR residual vectors, rebuild the series with the estimated VAR, re-estimate, recompute the table; 999 draws
  - ▶ Resampling whole residual vectors keeps the contemporaneous correlation  $\Sigma$ , which drives the generalized decomposition
- Full sample,  $p = 2$ :  $C = 63.9\%$  with 95% interval [59.9, 65.8]
  - ▶ Window of the peak (250 days to 2020-08-27,  $p = 2$ ):  $C = 79.3\%$ , interval [65.9, 83.0]; bootstrap mean 75.8%, so small samples also bias  $C$

## Net Spillovers with Bootstrap Intervals



- Full-sample VAR(2) on log Parkinson volatility,  $H = 10$ ; 95% intervals from 999 residual block-bootstrap draws (blocks of 20 days)

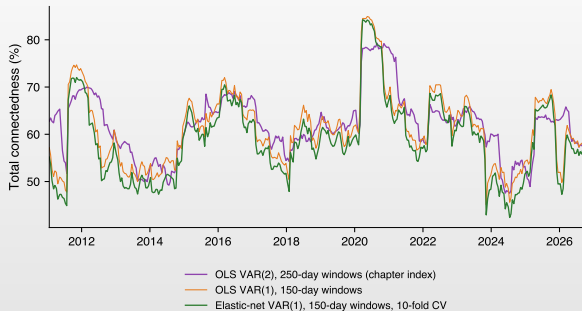
## Reading the Net Spillover Intervals

- 8 of 13 net positions have intervals that exclude zero; for the other 5 the label “transmitter” or “receiver” is not identified by the data
- The full sample has about 4 000 days; in the 250-day window of the 2020 peak the interval for  $C$  is almost three times wider ( $[65.9, 83.0]$ ), so rolling net labels are mostly noise
- Report the interval next to every net number, and draw only arrows whose pairwise net link is significant
- Frequency connectedness splits the same table into short- and long-horizon parts [Baruník & Křehlík \(2018\)](#); in their study of the global financial crisis, long-term connectedness gains importance after Lehman

## Regularised VARs: the Elastic Net

- [Demirer et al. \(2018\)](#), Eq. (9): each VAR equation by the adaptive elastic net,  $\lambda$  chosen equation by equation by 10-fold cross-validation
  - ▶  $\hat{\beta} = \arg \min_{\beta} \sum_t (y_t - \sum_i \beta_i x_{it})^2 + \lambda \sum_i w_i (\frac{1}{2} |\beta_i| + \frac{1}{2} \beta_i^2)$ ,  $w_i = 1/|\hat{\beta}_{i,OLS}|$
  - ▶ Rolling windows of 150 days with the plain elastic net,  $w_i = 1$  (Section 5, footnote 15); forecast horizon  $H = 10$
- Why: selection and shrinkage in the VAR coefficients, without forcing a sparse **network** (links also come through  $\Sigma$ )
  - ▶ The paper does not state the lag order in the text; we keep the BIC order of our OLS VAR,  $p = 2$
- Full sample, adaptive elastic net:  $C = 63.3\%$  (OLS 63.9%); 59% of the coefficients are non-zero
  - ▶ With about 4 000 days the penalty hardly matters; it matters in short windows (next slide)

## OLS and Elastic-Net Connectedness in Rolling Windows



- Total connectedness every 10 days,  $H = 10$ : the chapter index (OLS VAR(2), 250 days), OLS VAR(1) and elastic-net VAR(1) on 150-day windows [Demirer et al. \(2018\)](#)

## Reading OLS versus Elastic-Net Connectedness

- ▣ Means over 372 windows: 62.0% (chapter index), 61.1% (OLS, 150 days), 58.7% (elastic net, 150 days)
  - ▶ On the same 150-day windows OLS exceeds the elastic net by 2.4 points on average: estimation noise in the coefficients inflates measured connectedness
- ▣ The dynamics agree: correlation 0.82 between the elastic-net index and the chapter index; the crisis peaks are in the same weeks
- ▣ Levels depend on window, estimator and lag order; changes over time are the robust part of the index
- ▣ Long-run partial correlation networks for large panels: [Barigozzi & Brownlees \(2019\)](#)

## Granger-Causality Networks and Multiple Testing

- Billio, Getmansky, Lo & Pelizzon (2012) draw an edge  $j \rightarrow i$  when  $j$  Granger-causes  $i$  at 5%; with 13 banks this is  $13 \times 12 = 156$  tests
  - ▶ Even with no links at all, about 7.8 edges would appear by chance
- Our tests: bivariate regressions, one lag, heteroskedasticity-robust Wald statistics
  - ▶ Daily returns: 57 edges at 5%, 33 after Holm [Holm \(1979\)](#), 55 with the false discovery rate controlled at 5% [Benjamini & Hochberg \(1995\)](#)
  - ▶ Log volatility, highly persistent: one extra lag that is not tested [Toda & Yamamoto \(1995\)](#), as in the lag-augmented tests of [Shi, Hurn & Phillips \(2020\)](#): 153, 150 and 153 edges
- Holm controls the chance of any false edge under arbitrary dependence; the FDR is the **expected** share of false edges among those drawn, and BH guarantees it for independent tests [Benjamini & Hochberg \(1995\)](#), while tests on overlapping bank pairs are dependent
- Daily returns of banks in different time zones: many “causal” edges only reflect that New York closes after Frankfurt

## Recap: Spillover Networks

- ▣ The generalized variance decomposition of a VAR gives a directed, weighted network that does not depend on the ordering
- ▣ Two thirds of bank volatility risk comes from other banks; US banks transmit, HSBC and Romanian banks receive
- ▣ Connectedness jumps in crises and decays slowly: a monitoring tool for supervisors
- ▣ Choices matter: VAR order, horizon, window length, estimator and volatility versus returns (Seminar 18, B5)
- ▣ Every table and every edge has sampling error: bootstrap intervals, shrinkage, and multiplicity control for edge tests

## The Idea of the Financial Risk Meter

- For each firm  $j$ , a quantile regression at  $\tau = 5\%$  of its return on the returns of **all** other firms and on lagged macro variables [Mihoci, Althof, Chen & Härdle \(2020\)](#)
  - ▶ With 100 firms and a 63-day window there are more regressors than days: a penalty is needed
- **LASSO** (Least Absolute Shrinkage and Selection Operator) quantile regression [Belloni & Chernozhukov \(2011\)](#)
  - ▶  $\hat{\beta}_j(\lambda) = \arg \min_{\beta} \frac{1}{n} \sum_t \rho_{\tau} \left( X_{j,t} - \beta \mathbf{o} - \beta^{\top} Z_{-j,t} \right) + \lambda \|\beta\|_1$
  - ▶  $\lambda$  switches off weak links: a larger  $\lambda$  leaves fewer active coefficients
- The **FRM** (Financial Risk Meter) is the average of the selected  $\lambda_j$  over firms, designed as an indicator of tail interdependence and systemic stress
  - ▶ Built for US financial firms [Mihoci, Althof, Chen & Härdle \(2020\)](#), emerging markets [Ben Amor, Althof & Härdle \(2022\)](#) and China [Wang, Althof & Härdle \(2023\)](#)

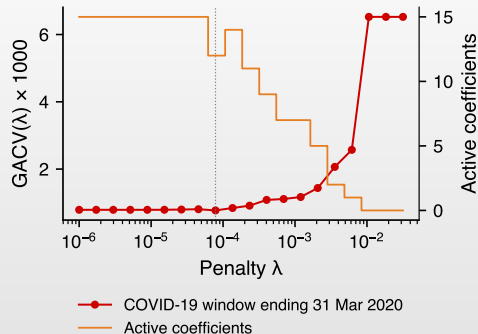
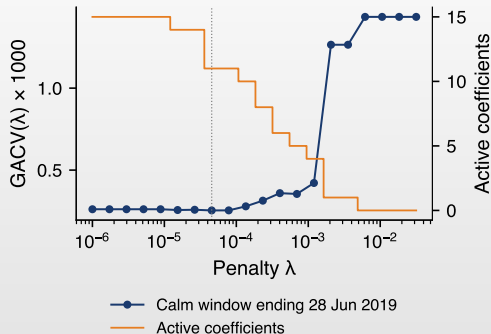
## Choosing the Penalty: GACV

- **GACV** (Generalized Approximate Cross-Validation) trades fit against the number of active coefficients

► 
$$\text{GACV}(\lambda) = \frac{\sum_t \rho_\tau \left( X_{j,t} - \hat{X}_{j,t}(\lambda) \right)}{n - \text{df}(\lambda)}, \text{ df} = \text{number of nonzero coefficients}$$

- $\hat{\lambda}_j = \arg \min_{\lambda} \text{GACV}(\lambda)$  on a grid;  $\text{FRM}_t = \frac{1}{J} \sum_j \hat{\lambda}_{j,t}$
- Our version: 13 banks, macro variables S&P 500, Euro Stoxx 50 and VIX at  $t - 1$ , 63-day windows every 10 days
  - 15 regressors for 63 days: far fewer than in the published FRM
- Extensions: FRM based on expectiles [Ren, Lu, Li & Härdle \(2022\)](#), for cryptocurrencies [Ren, Althof & Härdle \(2022\)](#) and for the EU energy sector [Toma, Dima, Grecu, Pele & Lessmann \(2026\)](#)

## GACV in a Calm and in a Crisis Window

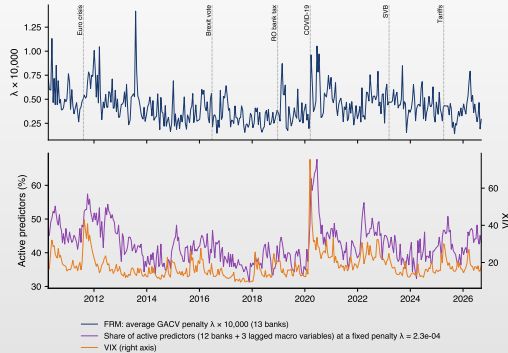


- JPMorgan regressed on the other 12 banks and three lagged macro variables, 63 days; dotted line: selected λ; orange: active coefficients

## Reading the GACV Curves

- Calm window (to 28 June 2019):  $\hat{\lambda} = 4.5 \times 10^{-5}$ , 11 of 15 coefficients active
  - ▶ COVID-19 window (to 31 March 2020):  $\hat{\lambda} = 7.8 \times 10^{-5}$ , 12 active
- The curve is flat for small  $\lambda$  and rises steeply once key links are switched off
- In the crisis the whole curve is about three times higher: the check loss is in return units, and returns are larger
- With few regressors, the minimum lies where almost no penalty is applied: the selection is noisy

## The Financial Risk Meter of Thirteen Banks



- Top: average GACV penalty over 13 banks; bottom: share of active predictors (12 banks and 3 lagged macro variables) when  $\lambda$  is held fixed, with the VIX; 390 windows of 63 days

## Reading Our Financial Risk Meter

- The average GACV penalty is noisy; its correlation with the VIX is only 0.35
  - ▶ With 15 regressors and 63 days the fit hardly needs a penalty; the published FRM uses about 100 firms, so that the penalty must do real work
- At a fixed penalty  $\lambda = 2.3 \times 10^{-4}$ , the share of active predictors (bank links and macro variables) follows the VIX more closely: correlation 0.62
  - ▶ From 31% (2017-11-21) to 68% (2020-06-17); average 42%
- Lesson: this small-panel application does not show that either the selected penalties or a fixed-penalty count of active predictors is a reliable systemic-risk indicator; the fixed-penalty count is only an exploratory sensitivity check

## Recap: the Financial Risk Meter

- Each firm's 5% quantile is regressed on all other firms with an L1 penalty; the penalty  $\lambda$  chosen by GACV summarises tail interdependence
- The FRM is designed to signal systemic stress; it relies on large cross-sections of firms
- On 13 banks the penalty is noisy; at a fixed penalty the share of active predictors co-moves with the VIX, without evidence that it predicts stress
- Advanced material: Quantinar course on the FRM (end of chapter)

## Climate Risk as a Systemic Risk

- ▣ **Physical risk:** losses from floods, droughts, heat and storms that damage collateral and borrowers
  - ▶ **Transition risk:** losses from the move to a low-carbon economy (carbon prices, regulation, technology)
- ▣ “Green swan” risks: potentially extremely disruptive events that could be behind the next systemic financial crisis; backward-looking risk models cannot anticipate them [Bolton et al. \(2020\)](#)
- ▣ A network climate stress test maps investors’ exposures to climate-policy-relevant sectors and adds second-round effects through the financial network [Battiston et al. \(2017\)](#)
- ▣ Hence scenario analysis instead of historical estimation: the past sample contains no energy transition

## The ECB Climate Risk Stress Test of 2022



Seat of the European Central Bank, Frankfurt, 2015

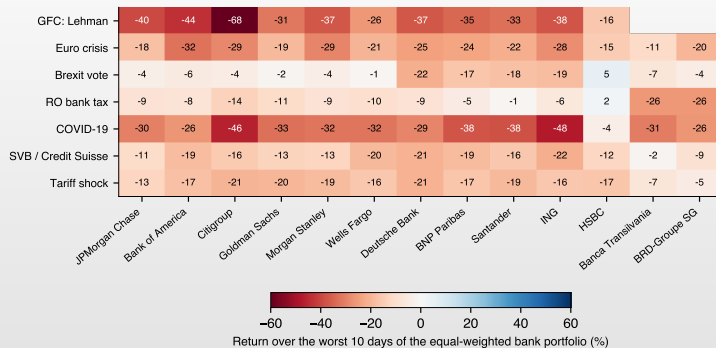
Photo: DXR (2015); CC BY-SA 4.0; Wikimedia Commons

- 104 significant banks took part; a bottom-up projection of losses covered 41 of them [ECB Banking Supervision \(2022\)](#)
- About 60% of banks had no climate risk stress-testing framework; only 20% considered climate risk when granting loans
- Almost two thirds of banks' income from non-financial corporate customers came from greenhouse gas-intensive industries
- Credit and market losses of around EUR 70 billion for the 41 banks, under a short-term disorderly transition and two physical risk scenarios
  - ▶ The ECB states that this significantly understates the risk: data gaps, rudimentary models, no second-round effects

## Stress Tests and Historical Scenarios

- A stress test asks how much we would lose in a **severe but plausible** scenario: **historical** (replay a past crisis), **hypothetical** (shock chosen factors), **reverse** (find the most plausible scenario behind a fatal loss)
  - ▶ Supervisory tests of large banks: [Federal Reserve stress tests](#) in the US, EU-wide tests of the [European Banking Authority](#); global systemically important banks hold extra capital [Basel Committee \(2018\)](#)
- Seven episodes, each a window of a few weeks around a known shock
  - ▶ Lehman (autumn 2008), the euro crisis (summer 2011), the Brexit vote (June 2016), the Romanian bank tax (December 2018), COVID-19 (March 2020), SVB and Credit Suisse (March 2023), the tariff shock (April 2025)
- Inside each window we find the **worst 10 trading days** of the equal-weighted bank portfolio and report every bank's return over those days
- The 2008 episode uses the 11 international banks only: the Romanian series start in 2010
- Ten days match the horizon of market-risk capital (Chapter 7)

## Seven Historical Stress Scenarios



Return (%) of each bank over the worst 10 days of the equal-weighted bank portfolio in each episode; blank: no data

## Reading the Historical Scenarios

- Portfolio loss over the worst 10 days: Lehman 37.6% (2008-11-10 to 2008-11-21), COVID-19 31.9%, euro crisis 22.3%
  - ▶ Tariffs 16.0%, SVB 14.8%, Romanian bank tax 10.2%, Brexit 7.9%
- Worst single cells: Citigroup −68% in 2008, ING −48% in March 2020
- Shocks are local as well as global: Brexit hit mainly European banks; the Romanian bank tax hit Romanian banks hardest (−26% and −26%), during a December 2018 sell-off that also cost US banks −8% to −14%
- A stress test built only on global crises would miss the domestic political shock

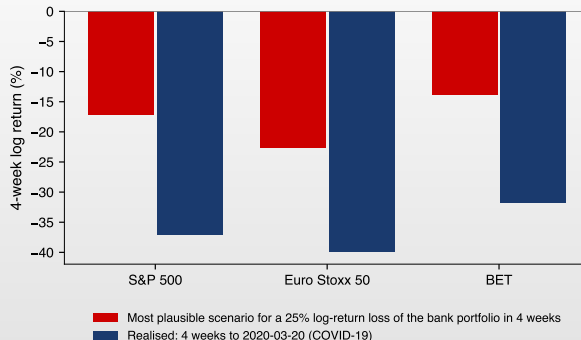
## Reverse Stress Testing

- Factor model: weekly bank log returns (%)  $X_i = \beta_i^\top f + e_i$ , factors  $f = (\text{S\&P 500, Euro Stoxx 50, BET})$ , over  $h$  weeks  $f \sim N(0, h\Sigma)$  (no serial covariance of weekly factor returns)
  - ▶ Loss of the equal-weighted portfolio in log-return points:  $L = -b^\top f$ , with  $b$  the average of the  $\beta_i$ ; a factor-only approximation that sets the residual  $\sum_i w_i e_i$  to zero
- The most plausible scenario with loss  $\ell^*$  minimises the Mahalanobis distance  $f^\top (h\Sigma)^{-1} f$  subject to  $-b^\top f = \ell^*$ 
  - ▶  $f^* = -\ell^* \frac{h\Sigma b}{b^\top h\Sigma b}, \quad d^* = \frac{\ell^*}{\sqrt{b^\top h\Sigma b}}$
- $d^*$  counts portfolio standard deviations: the larger it is, the less plausible the loss under the model

## Worked Example: Reverse Stress Test with Two Factors

- Betas  $b = (1.2, 0.8)$ ; four-week covariance of the factors  $h\Sigma = \begin{pmatrix} 16 & 6 \\ 6 & 9 \end{pmatrix}$  (%<sup>2</sup>); target loss  $\ell^* = 20\%$ 
  - ▶  $h\Sigma b = (1.2 \cdot 16 + 0.8 \cdot 6, ; 1.2 \cdot 6 + 0.8 \cdot 9) = (24, 14.4)$
  - ▶  $b^\top h\Sigma b = 1.2 \cdot 24 + 0.8 \cdot 14.4 = 40.32$ , standard deviation 6.35%
- Most plausible scenario and its distance
  - ▶  $f^* = -20 \cdot (24, 14.4)/40.32 = (-11.9\%, -7.1\%)$ ,  $d^* = 20/6.35 = 3.15$
  - ▶ Normal probability of a loss at least this large:  $\Phi(-3.15) = 0.08\%$  per four weeks
- The scenario loads more on the factor with the larger beta and variance, and both factors fall together because they are correlated

## The Most Plausible Scenario for a 25% Log-Return Loss



- Weekly data 2010–2026 (871 weeks), four-week horizon; red:  $f^*$  for a log-return loss  $\ell^* = 25\%$ ; blue: factor moves in the worst four weeks of 2020

## Reading the Reverse Stress Test

- Portfolio betas:  $b = (0.45, 0.65, 0.18)$ ; four-week standard deviation 5.79%
  - ▶ A log-return loss of 25% (an arithmetic loss of 22.1%) needs log returns of S&P 500  $-17.2\%$ , Euro Stoxx 50  $-22.6\%$ , BET  $-13.9\%$  together
- Distance  $d^* = 4.31$ : under the Normal model its probability is  $8.0 \times 10^{-6}$  per four weeks, practically impossible
- Yet in the four weeks to 2020-03-20 the portfolio lost 52.0% in log terms, with factor falls larger than  $f^*$  (the model implies 48.5%)
- The shape of the scenario is informative; its probability is not: heavy tails and rising correlations make Normal probabilities far too small

## Limits of Stress Tests

- ▣ Scenarios are chosen by people: the crisis that comes is rarely the one that was tested
- ▣ Second-round effects (fire sales, funding runs, counterparty losses) are hard to model and usually left out, as the ECB noted for its climate test
- ▣ Market-based measures (MES, CoVaR, connectedness) react quickly but only after prices move; balance-sheet stress tests look forward but slowly
- ▣ Early-warning research combines both, e.g. entropy-based measures of market risk [Pele, Lazar & Dufour \(2017\)](#)

## Recap: Stress Testing

- ▣ Historical, hypothetical and reverse stress tests answer different questions
- ▣ Across seven episodes, the worst 10-day loss of a bank portfolio ranged from 7.9% to 37.6%
- ▣ Reverse stress tests give the most plausible direction of a disaster, not its probability
- ▣ Climate scenarios extend stress testing to risks that have no history

## Case Study: Question and Motivation

- Ando, Greenwood-Nimmo & Shin (2022) ask whether the topology of a financial network changes with the **size** of the shocks that hit it (Section 1)
  - ▶ Their version of the question of Diebold & Yilmaz (2014): how much of the future uncertainty of  $i$  is due to idiosyncratic shocks from  $j$ , as the shock size varies?
- Why it matters: the connectedness of this chapter is estimated at the conditional mean (OLS, LASSO, elastic net Demirer et al. (2018)), i.e. it describes how an average shock spreads
  - ▶ Systemic shocks are larger than average; reading a rise in rolling connectedness as a crisis signal assumes that mean relations also hold in the tails (Section 1)
- Link to this chapter: the quantile regression of CoVaR Adrian & Brunnermeier (2016), Koenker & Bassett (1978), now inside the VAR and its variance decomposition
  - ▶ Giglio, Kelly & Pruitt (2016) showed that systemic risk shifts the lower tail of future output more than its centre; Ando, Greenwood-Nimmo & Shin (2022) ask the same tail-versus-centre question of the network itself
- Application: the credit risk loop between sovereigns and their banks, in which bailouts move bank risk onto the state (Sections 1 and 3)

## Case Study: Data and Method as Published

- Data (Section 3): daily first differences of 5-year CDS spreads (basis points) of 17 sovereigns and 17 national financial sectors,  $m = 34$ , 3 January 2006 – 29 December 2017, 3 129 days
  - ▶ A financial sector = equal-weighted CDS of its banks and insurers with assets of at least USD 10 billion, 199 institutions in all
- Quantile factor VAR (QFVAR), Eqs. (5)–(7):  $y_t = \mu + \sum_{j=1}^p \Phi_j y_{t-j} + \Lambda f_t + v_t$ , each equation fitted at quantile  $\tau$  by minimising the check loss
  - ▶ Unobserved factors  $f_t$  absorb common shocks, so  $v_t$  is idiosyncratic; their number by the criterion of [Ando & Bai \(2020\)](#), Eq. (8): 10 at the mean (Figure 1)
  - ▶ Residual pairs with  $|\rho| < 0.25$ : 91.6% with factors, fewer than half without (Figure 2)
- Generalized FEVD of the idiosyncratic shocks at quantile  $\tau$ , Eq. (17), rows normalised to 1; spillover index  $S_\tau^{(h)} = m^{-1} \sum_i F_{i \leftarrow \bullet, \tau}^{(h)}$ , Eq. (20)
  - ▶ VAR(1) chosen by BIC at the mean and kept at every quantile;  $h = 5$  days; the quantile is held fixed along the forecast horizon (Sections 2.2 and 4)
- Rolling windows of  $w = 250$  days (2 880 windows), number of factors fixed at its full-sample value; relative tail dependence
 

RTD =  $S_{0.95}^{(5)} - S_{0.05}^{(5)}$  (Section 4.3)

  - ▶ For CDS changes the right tail ( $\tau = 0.95$ ) holds large adverse shocks, the left tail ( $\tau = 0.05$ ) large beneficial ones

## Case Study: Key Results

- ▣ Full-sample spillover index (Figure 4): about 40% over the central 40% of the distribution; 72.20%, 77.17%, 88.18% at  $\tau = 0.10, 0.05, 0.01$ ; 73.34%, 79.36%, 91.77% at  $\tau = 0.90, 0.95, 0.99$ ; 56.57% at the mean
  - ▶ Small shocks stay local; large shocks, good or bad, spread; the mean averages the two regimes into one number
- ▣ Of the 1122 bilateral spillovers, 568 are below 1% at the mean and 829 at the median (Figure 7)
  - ▶ Away from the mean, the network has many weak links rather than a few strong ones; in the tails these weak links add up to strong average spillovers
- ▣ Rolling windows (Table 3): the mean index correlates 0.774 with  $S_{0.95}$  but 0.411 with  $S_{0.05}$ ; the two tails correlate only 0.367
  - ▶ The tails move apart: RTD is 2.19% on the full sample but reaches about  $\pm 15\%$  in rolling windows (Section 4.3, Figure 11)
- ▣ Monthly correlation with CATFIN Allen, Bali & Tang (2012), the average of three VaR 1% estimates for financial stocks (Figure 13): 0.51 for  $S_{0.95}$ , 0.15 for  $S_{0.05}$ , 0.42 for RTD
  - ▶ RTD rises with the Bear Stearns fund losses of 2007 and falls after the US bank rescue programme and the negative rates of the ECB: a fragility signal the mean hides

## Case Study: Replicate, Limits, Extend

- Replicate with the course data: daily closes of the 13 banks of this chapter in data/market (US, euro area and United Kingdom, Romania), 2010–2026;  $y_t$  = daily loss  $-r_t$  in %, so that the right tail is the adverse one, as for CDS
  - ▶ Specification as published: VAR order by BIC, factors by Eq. (8),  $h = 5$ ,  $\tau \in \{0.05, 0.5, 0.95\}$  and the mean,  $w = 250$ , RTD
  - ▶ Data limitation to state: equity losses instead of CDS spreads, 13 banks and no sovereigns ( $m = 13$  instead of 34), different closing times of US, European and Romanian markets
- Limits stated by the authors: the lag order of the mean is imposed at every quantile, and the quantile is held fixed along the horizon (Sections 2.1–2.2, footnote 6)
  - ▶ 250 days are too few to select factors window by window, so their number is fixed (Section 4.3, Appendix B); shorter windows leave too few effective tail observations
  - ▶ Only direct spillovers of idiosyncratic shocks are measured, not propagation through the common factors (footnote 2); Greece is left out for a gap in its CDS data (Section 3, Appendix A)
- Open extension: does the RTD of the 13 banks rise before March 2020 and the March 2023 bank failures, and does it predict bank stress beyond current volatility and the mean-based index of this chapter?
- Numbers from the accepted manuscript of 19 October 2020, [University of Melbourne repository](#)

## An Open Question

### Does network connectedness measure systemic risk, or only volatility?

- Connectedness indices are monitored as early-warning tools, but an index built from a VAR on volatilities may co-move with volatility without adding predictive information
  - ▶ Live research: global bank networks from regularised VARs [Demirer et al. \(2018\)](#); connectedness in the tails differs from connectedness in the mean [Ando, Greenwood-Nimmo & Shin \(2022\)](#)
- Formal version, for future bank stress  $y_{t+h}$ , total connectedness  $C_t$  and current volatility  $\sigma_t$ :  
$$y_{t+h} = a + \beta_C C_t + \gamma \ln \sigma_t + e_{t+h}, \quad H_0: \beta_C = 0, \quad H_1: \beta_C > 0$$
  - ▶ Falsified if  $C_t$  loses its predictive power once current volatility is controlled
- Identification: a network edge is a share of forecast error variance, not a causal exposure
  - ▶ A common shock raises volatility and connectedness together, with no bank infecting another

## The Discovery Loop, Step by Step (1/2)

- **Literature:** map what is known, then verify every reference
  - ▶ “Which papers test whether connectedness indices predict bank distress out of sample, beyond volatility? Give DOI, sample, horizon and controls.”
  - ▶ Resolve each DOI on [Crossref](#) and check that the paper really controls for volatility; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses:** competing readings of a rise in connectedness, each with its own test
  - ▶ “Give competing interpretations of a rise in total connectedness (common shock, contagion, artefact of rolling windows) and a test that discriminates between them.”
  - ▶ E.g. a common shock fades with volatility; contagion has a direction: shocks of net transmitters precede those of receivers
- **Data and code:** on the course data, replicating a known number first
  - ▶ “Fit a VAR(2) to the log Parkinson volatilities of the 13 banks in data/market on 250-day windows and compute the generalized total connectedness with  $H = 10$ .”
  - ▶ Replicate first: mean 62.0% and maximum 79.3% (2020-08-27), as in this chapter

## The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed before the predictive regressions are run
  - ▶ “Pre-register: horizons 21/63/126 days, volatility and VIX controls, windows 150/250/500 days, HAC lags at least the overlap, out-of-sample evaluation from 2018.”
  - ▶ Every cell of the grid is reported; the family is corrected with Holm or controlled with the FDR
- **Critique**: ask the assistant to attack the result
  - ▶ “Act as a referee: is the predictive power of connectedness just the persistence of volatility? Which regression would show it?”
  - ▶ Expect: add current volatility as a control, then test the increment of  $R^2$  and of the out-of-sample loss
- **Report**: standardised effects with HAC  $t$ -statistics, every cell of the grid, what failed
  - ▶ The prompts that shaped the design go into AI\_USE.md

## What the Researcher Must Check

- **Non-stationary inputs:** a VAR on price levels, or on trending series, produces spurious connectedness
  - ▶ Test: unit-root tests on every input; all eigenvalues of the companion matrix of the fitted VAR strictly inside the unit circle in every window
- **Ordering:** Cholesky decompositions change with the order of the banks
  - ▶ Test: permute the columns; the generalized table must not change
- **Look-ahead:** a window labelled by its first date, or a lag order chosen on the full sample
  - ▶ Test: the index at  $t$  must not move when data after  $t$  are deleted
- **Overlap:** rolling indices and forward outcomes overlap, so OLS  $t$ -statistics are inflated
  - ▶ Test: Newey–West lags at least equal to the overlap, or a block bootstrap
- **Dimension:** 13 banks and  $p = 2$  give 27 coefficients per equation from 250 days; more banks need shrinkage [Demirer et al. \(2018\)](#)
  - ▶ Test: compare with a LASSO VAR; unstable edges signal estimation noise, not structure

## Mini-Case: Is Connectedness More Than Volatility?

- A typical assistant proposal: “high total connectedness warns of future bank stress”; we test it on 731 weekly dates, 2011–2026
  - ▶  $y_{t+h}$ : log mean Parkinson volatility of the 13 banks over the next  $h = 63$  days; control: the same over the last 20 days; standardised variables, Newey–West with 13 lags
- Connectedness alone:  $\hat{\beta}_C = 0.40$  ( $t = 4.63$ ),  $R^2 = 0.16$ : the proposal looks right
- With current volatility:  $\hat{\beta}_C = 0.08$  ( $t = 0.75$ ),  $\hat{\gamma} = 0.53$  ( $t = 4.13$ );  $R^2 = 0.343$  vs 0.339 with volatility alone
  - ▶ Correlation of  $C_t$  with current volatility: 0.59; for the next 63-day maximum drawdown of the bank portfolio,  $\hat{\beta}_C = -0.10$  ( $t = -1.32$ ) even alone
- Interpretation: once current volatility is controlled, the increment of connectedness is not significant; this does not prove it is zero, but in our sample the persistence of volatility accounts for most of the predictive content; network information may still matter in the tails [Ando, Greenwood-Nimmo & Shin \(2022\)](#) or for the direction of transmission

## Your Turn: a Project Seed

- **Question:** does **tail** connectedness predict bank distress beyond volatility, out of sample?
- **Replicate first:** our rolling index (mean 62.0%, spring 2020 78.4%), then one headline connectedness figure of [Demirer et al. \(2018\)](#) on their sample
  - ▶ Extend: quantile connectedness at  $\tau = 5\%$  [Ando, Greenwood-Nimmo & Shin \(2022\)](#), out-of-sample evaluation from 2018, MES and SRISK as alternative targets
- **Deliverables:** a pre-registered grid committed before the regressions, a notebook that regenerates every number, HAC inference and out-of-sample losses for every cell
- **AI log:** prompts that shaped the design in AI\_USE.md; each wrong claim (a VAR on price levels, OLS  $t$ -statistics on overlapping windows) and the test that caught it in AI\_ERRORS.md ([templates](#))

## Which Bank Is Most Systemic?

| Bank               | VaR 1% | MES 5% | $\Delta\text{CoVaR 1\%}$ | Net spillover | $L^*$ |
|--------------------|--------|--------|--------------------------|---------------|-------|
| JPMorgan           | 4.83   | 3.22   | 4.62                     | 9.8           | 7.0   |
| Bank of America    | 5.99   | 3.85   | 4.60                     | 10.9          | 6.2   |
| Citigroup          | 5.87   | 4.03   | 3.81                     | 19.0          | 6.0   |
| Goldman Sachs      | 4.81   | 3.34   | 4.12                     | 2.4           | 6.8   |
| Morgan Stanley     | 5.83   | 4.04   | 4.00                     | 9.9           | 6.0   |
| Wells Fargo        | 5.78   | 3.32   | 4.89                     | 3.6           | 6.8   |
| Deutsche Bank      | 7.19   | 4.54   | 4.24                     | -12.6         | 6.1   |
| BNP Paribas        | 6.18   | 4.53   | 4.84                     | 6.0           | 6.1   |
| Santander          | 5.83   | 4.23   | 4.42                     | -5.4          | 6.4   |
| ING                | 7.16   | 4.81   | 4.42                     | 5.2           | 5.8   |
| HSBC               | 4.66   | 2.54   | 4.07                     | -23.6         | 8.3   |
| Banca Transilvania | 5.06   | 3.14   | —                        | -12.2         | 6.5   |
| BRD-Groupe SG      | 4.42   | 2.69   | —                        | -13.0         | 7.1   |

□ Daily losses in %; net spillover of volatility (to minus from);  $L^*$ : break-even leverage with  $k = 8\%$

## Reading the Comparison

- ▣ Rank correlations across the eleven international banks
  - ▶ MES and  $\Delta\text{CoVaR}$ :  $-0.01$ ; MES and net spillover:  $0.04$ ;  $\Delta\text{CoVaR}$  and net spillover:  $-0.01$
  - ▶ VaR and MES:  $0.88$
- ▣ The measures answer different questions: exposure to the market, contribution to the system, transmission of volatility
- ▣ No single ranking: supervisors combine several indicators with size and balance-sheet data, as in the [Basel Committee \(2018\)](#) methodology
- ▣ Romanian banks: moderate MES, net receivers of volatility, break-even leverage close to that of US banks

## Key Takeaways

- ▣ Systemic risk lives in the joint tail: exposure (MES, SRISK), contribution ( $\Delta\text{CoVaR}$ ) and connectedness (spillover and tail networks)
- ▣ European banks have the largest MES in our sample; US banks are the main net transmitters of volatility
- ▣ VaR and  $\Delta\text{CoVaR}$  rank banks differently; rankings are uncertain: test differences, report intervals, state  $\alpha n$
- ▣ Market-based measures jump in crises; stress tests and balance sheets are needed to see risk building up
- ▣ Romanian banks are mostly driven by domestic shocks and receive more than they transmit

## Common Pitfalls

- ▣ Reading  $\Delta\text{CoVaR}$  as causal: a common shock creates the same pattern
- ▣ Reading overlapping intervals as “equal” and separate ones as “different”: test the difference on joint draws, and correct for the number of pairs
- ▣ Including the bank itself in the system portfolio: its own return adds a mechanical self-contribution to the link
- ▣ Orthogonal variance decompositions whose results depend on the order of the banks
- ▣ Reporting SRISK without balance-sheet data, or from the  $1 - e^{-18\text{MES}}$  shortcut as if it were the published measure
- ▣ Drawing every Granger edge significant at 5% out of 156 tests, or every net label without its interval

## Further Study on Quantinar

### Advanced courses on quantinar.com

- ▣  Financial Risk Meter for Emerging Markets
- ▣  Dynamic Crypto Networks
- ▣  Network Centrality

### Code and data

- ▣ Quantlets: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_18](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_18)

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